



RUDRA SAHU

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RESEARCH INTERESTS

My current research is broadly focused on the interface between gravitational-wave astronomy and the optical systems that enable these observations. I am also drawn to computational astrophysics, numerical simulations, and data-driven approaches to studying astronomical phenomena, from stellar systems to the large-scale Universe.

Keywords: Gravitational-Wave Astronomy, Computational Astrophysics, Scientific Computing, Numerical Simulations, Data Analysis.

EDUCATION

Indian Institute of Science Education and Research (IISER), Pune, India

August 2024 - present

Five year BS-MS Dual Degree Programme

DAV Model School, IIT Kharagpur

April 2012 - May 2024

Secondary and Higher Secondary in PCM stream (CBSE)

RESEARCH EXPERIENCE

Investigating Interferometric Techniques for Gravitational-Wave Detection

May 2026 - present

Supervisor: Dr. Deepak Pandey, IUCAA, Pune, India

- Investigating the optical cavities and interferometry used in gravitational-wave observatories such as LIGO, with a focus on understanding the physics behind precision measurements.
- Working through the design and behaviour of resonant optical systems, exploring how different cavity configurations affect beam properties and system performance.
- Building simulations using Finesse 3 to study cavity dynamics, model interferometric systems, and develop intuition for the techniques used in modern gravitational-wave detection experiments.

AWARDS & SCHOLARSHIPS

INSPIRE - DST Scholarship

August 2024 - present

Awarded by the Department of Science and Technology in recognition of academic excellence and potential for pursuing a career in scientific research.

Reliance Foundation Undergraduate Scholarship

August 2024 - present

Awarded by the Reliance Foundation to support undergraduate studies, recognising academic achievement, leadership potential, and commitment to personal and professional growth.

TECHNICAL AND SOFTWARE SKILLS

Scientific Computing & Programming: Experience using Python, numerical simulations, data analysis, and scientific computing workflows.

Optical Modelling & Simulation: Working knowledge of Finesse 3 for modelling optical cavities and interferometric systems, with applications to gravitational-wave detection and precision measurements

Data Analysis & Research Computing: Proficient in using NumPy, Pandas, Matplotlib, and Astropy for analysing scientific datasets, visualisation, and computational research.

CODING SKILLS

Languages: Python, L^AT_EX

Libraries: NumPy, Pandas, Matplotlib, Astropy, Sunpy, Finesse 3

CONFERENCES AND WORKSHOPS

Astronomical Hub International Summer School for Astronomy (AHISSA '26)

July 2026

Astronomical Hub (ISRO Registered Space Tutor)

- Participated in an 8-day online international summer school covering astronomy, astrophysics, cosmology, black holes, astrophotography, space weather, light pollution, extraterrestrial life, and science diplomacy.
- Attended expert lectures by researchers and science communicators from institutions including the Max Planck Institute for Radio Astronomy (Germany), ARIES, and Université Paris Cité.
- Gained exposure to contemporary topics in theoretical astrophysics, observational astronomy, scientific outreach, and interdisciplinary applications of astronomy.

Intro2Astro 2025

July 2025 - August 2025

University of Hawai'i, UC Berkeley, and ZS Associates

- Gained hands-on experience with Python and key scientific libraries (NumPy, Pandas, Matplotlib, Astropy, SciPy), LaTeX typesetting, basic web development for scientific outreach, and academic CV creation.
- Engaged in extensive tutorials on exoplanet detection methods, exoplanetary atmospheric science, use of the NASA Exoplanet Archive, and TESS mission data.

Winter School in Astronomical Calculations using Python Programming

December 2024 - January 2025

Wilson College and Indian Association Of Physics Teachers (IAPT)

- Received hands-on training in computational astronomy, covering heliophysics, stellar and galactic astrophysics, exoplanet detection, cosmology, and multi-messenger astronomy through data-driven projects in Python.
- Worked with astronomical datasets using Astropy, Pandas, and SunPy to analyse phenomena including stellar clusters, Cepheid variables, X-ray binaries, black-hole jets, solar observations, and Hubble-law measurements, developing practical experience in astronomical data analysis and modelling.

POSITIONS OF RESPONSIBILITIES AND OUTREACH

• HELICASE Coordinator, The Science Club, 2025–26

Worked with the editorial team to design and put together HELICASE, the Science Club magazine. Oversaw the magazine's aesthetic direction and publication design.

• Design & Decoration Coordinator, Mimamsa, 2025–26

Led the conceptualisation and execution of visual design and event aesthetics. Managed the visual identity and event decor, coordinating design elements across venues and promotional materials to create a cohesive experience.